Exponentiation II

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1712 | Mathematics | https://cses.fi/problemset/task/1712

Your task is to efficiently calculate values a^{b^c} modulo 10^9+7.

Note that in this task we assume that $0^0=1$.

Input

The first input line has an integer n: the number of calculations.

After this, there are n lines, each containing three integers a, b and c.

Output

Print each value a^{b^c} modulo 10^9+7.

Constraints

- $1 \le n \le 10^5$
- $0 \le a,b,c \le 10^9$

Example

Input

```
3
3 7 1
15 2 2
3 4 5
```

Output

```
2187
50625
763327764
```